EUNJEONG PARK

Ph.D student in Mechanical Engineering, The Pennsylvania State University
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RESEARCH INTERESTS

My research interests lie in AI-driven physics-based modeling for materials and manufacturing systems. I am particularly interested in integrating machine learning, physics-informed neural networks, large language models, and diffusion models to understand, predict, and optimize compositional, microstructural, and structural behaviors in additive manufacturing and advanced materials.

EDUCATION

The Pennsylvania State University, United States *Aug.2025 - May.2029*

- Ph.D in Mechanical Engineering, GPA: 3.9/4.0
- Advisor: Prof. Amrita Basak
- Thesis: AI-enabled Compositional and Microstructure Evaluation

Korea Advanced Institute of Science and Technology, South Korea *Sep.2020 - Aug.2022*

- MS in Mechanical Engineering, GPA: 3.9/4.3
- Advisor: Prof. Seunghwa Ryu
- Thesis: Machine Learning-based Elastic Modulus Prediction and Structural Shape Optimization

Inha University, South Korea *Feb.2017 - Aug.2020*

- BS in Mechanical Engineering, GPA: 4.3/4.5 (Early Graduation)

EXPERIENCE

Aug.2022–Jul.2025	**R&D Engineer**, *Samsung Electronics Semiconductor R&D Center*, South Korea.
Aug.2020–Aug.2022	**Graduate Research Assistant**, *KAIST*, South Korea.
Mar.2020–Jul.2020	**Student Assistant**, *3D Printing & Laser Cutting Center, Inha University*, South Korea.
Sep.2019–Dec.2019	**Intern**, *KIA Motors*, South Korea.

HONORS & AWARDS

Aug.2025	**University Graduate Fellowship**, *The Pennsylvania State University* – Awarded to outstanding incoming Ph.D. students in recognition of academic excellence and research potential.
Mar.2020	**Hanjin Foundation Scholarship**, *Hanjin Group* – Selected as a recipient of a competitive scholarship supporting high-achieving undergraduate students with strong academic performance.
Feb.2020	**Super Challenge Hackerton, Best Award**, *Inha Univ. Start-up Center.*
Feb.2019	**University Simulation Competition, Honorable Mention**, *MSC Software.*
Oct.2018	**3D Printing Creative Competition, Grand Prize**, *KSPSE.*
Sep.2018	**Comprehensive Design Competition, Grand Prize**, *Inha Univ. Innovation Center.*

JOURNAL PAPERS

[2] **Eunjeong Park** and Amrita Basak. "Multi-Label Phase Diagram Prediction in Complex Alloys via Physics-Informed Graph Attention Networks." *arXiv preprint arXiv:2604.16468* (2026).

[1] Hyunggwi Song, **Eunjeong Park**, Hong Jae Kim, Chung-Il Park, Taek-Soo Kim, Yoon Young Kim, and Seunghwa Ryu. "Free-form optimization of pattern shape for improving mechanical characteristics of a concentric tube." *Materials & Design* 230 (2023): 111974.

CONFERENCE PRESENTATIONS

[4] **Eunjeong Park** and Amrita Basak. "Out-of-Distribution Phase Equilibrium Prediction Using Physics-Informed Graph Attention Networks." *2027 TMS Annual Meeting & Exhibition*, AI/ML/Data Informatics for Materials Discovery Symposium, **Accepted for Oral Presentation**, 2027.

[3] **Eunjeong Park** and Amrita Basak. "Physics-Informed Graph Attention Networks for Accurate Multi-Phase Prediction in Ag–Bi–Cu–Sn Alloys." *College of Engineering Research Symposium (CERS), The Pennsylvania State University*, 2026.

[2] **Eunjeong Park** and Seunghwa Ryu. "Machine Learning-based Pattern Shape Optimization to Improve the Mechanical Performance of Concentric Tube." *The Korean Society of Mechanical Engineers Annual Conference*, 2022.

[1] Jiyoung Jeong, Yongtae Kim, **Eunjeong Park**, and Seunghwa Ryu. "Transfer Learning Framework Utilizing Homogenization Theory to Predict the Effective Behavior of Elasto-Plastic Composites." *The Korean Society of Mechanical Engineers Annual Conference*, 2021.

RESEARCH MENTORING

High School Student Research Mentoring: Juliet Rueda *Mar.2026 - Present*

- Mentoring Juliet Rueda, an 11th-grade student, on a research project developing a machine learning framework to predict material properties of Al_2O_3–Cu composites
- Guiding feature design using phase-field maps and grain-level microstructural information for structure–property prediction
- Advising on research framing, poster organization, and scientific communication; the project received third place in the Somers High School Science Research Fair poster category

REVIEWING ACTIVITIES

[1] **Peer Reviewer**, *Engineering Applications of Artificial Intelligence (Elsevier)*, 2026.

LANGUAGES

Korean (Native) **English** (Working Proficiency)

ACADEMIC SKILLS

Programming Languages	Python, MATLAB
Simulation & Analysis	COMSOL Multiphysics, Ansys, ADAMS, SPPARKS, CALPHAD
CAD	Solid Edge, NX, AutoCAD
AI	Machine Learning, Deep Learning, Gaussian Process, LLM, Vision Transformer (ViT), Diffusion Model
Fabrication & Optics	3D Printing (Metal/Polymer), Microscopy, AFM, e-Beam, WLI